

PRE-ORAL COMPLIANCE REPORT

TITLE: QUEUEX: A COMPUTER VISION-BASED SMART QUEUE MANAGEMENT AND PREDICTIVE SERVICE DELIVERY SYSTEM

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SUGGESTIONS/ RECOMMENDATIONS	ACTION TAKEN	PAGE NUMBER
New Title: A Computer Vision-Based Smart Queue Management and Predictive Service Delivery System	Adopted the new title, preceded by the system name QueuEx, on the title page and throughout the manuscript.	Title page
Preliminaries: Preliminaries are required during the Pre-Oral Defense.	Included the title page, approval sheet, certification pages, acknowledgment, abstract, table of contents, list of figures, and list of tables.	Preliminary pages
Introduction: Indent the first paragraph. Use the funnel approach (global situation to national situation down to local, where the study is conducted).	Indented the first paragraph. The Background of the Study now moves from queue management as a global service-delivery problem, to Philippine higher education institutions, to Naga College Foundation, where the study is conducted.	1

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Review of Related Literature: 30 RRLs per topic/theme.	Organized the review into four themes, each with at least 30 studies: person detection (35), face recognition and open-set identification (30), queue management and re-identification (32), and predictive analytics for service demand (32).	3–13
Synthesis of the State-of-the-Art: Indent the first paragraph.	Indented the first paragraph.	13
Gap Bridged by the Study: Indent the first paragraph. Should summarize the features of the system presented in the objective of the study.	Indented the first paragraph and summarized the system's features in the order of the objectives: the baseline survey, face-based identification and queue-number issuance, wait-time prediction measured by MAE and MAPE, and user satisfaction.	14
Theoretical Framework: Discuss the main theory and the supporting theories. Use the recent citations of the theories used. Forecast and predictive theories are outdated; 2004 and 2011 are already outdated.	Presented Queuing Theory as the main theory, with Computer Vision, Deep Face Recognition, and Forecasting Theory as supporting theories. The 2004 and 2011 citations were replaced by Hyndman and Athanasopoulos (2021), and recent citations were added for each theory (2021–2023).	14–17
Theoretical Paradigm: Make it bigger and readable.	Redrawn at full page width, with the main theory above the system and the three supporting theories below it.	15
Conceptual Paradigm: Make it bigger and readable.	Redrawn at full page width as an Input–Process–Output–Outcome diagram with a feedback loop, updated to the current system.	17

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Statement of the Problem: The four research problems given by the panel.	Adopted the four research problems. Problem No. 2 refers to the face-recognition system rather than a YOLO-based system, following the panel's recommendation to use an appropriate algorithm for facial recognition: YOLO detects persons, and face recognition identifies them.	18
Scope and Delimitation: Order of discussion: 1) introduction, 2) scope, 3) limitation.	Presented in the required order: introduction, scope, and limitations.	19
Definition of Terms: Indent an introductory statement before defining the terms used in the study.	Added an indented introductory statement before the terms.	20
Population and Locale: Rephrase the first paragraph, directly addressing NCF as the locale of the study. How many will be considered as the population? And who are they?	Rephrased the first paragraph to address NCF directly as the locale, and identified the population as three groups: the cashiering-office staff, the administrators and IT personnel who oversee the office, and the students who transact there. The number in each group is to be confirmed with NCF administration before deployment.	24
Sample Technique: What is the sampling technique used?	Stated the techniques used: total enumeration for staff and administrators, stratified purposive sampling guided by Slovin's formula for students, and convenience sampling for the baseline survey.	25
System Architecture: Make it readable.	Redrawn at a larger scale as six labelled layers, updated to the current system, and marked by where each layer runs.	35

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Software Development Process: Figure 2.3: do not include unless you personalize or customize the content based on the study's stages and phases.	Removed Figure 2.3.	40
Validation and Testing Methods: If you have a researcher-made questionnaire, include it in the discussion of reliability testing.	Added reliability testing with Cronbach's alpha for both researcher-made questionnaires. The baseline questionnaire's results are reported in Table 3.1.	42
Results and Discussion: Does not align with your SOP: 3.1 SOP No. 1, 3.2 SOP No. 2, 3.3 SOP No. 3.	Restructured Chapter 3 so that each section answers one research problem in order: 3.1 SOP No. 1, 3.2 SOP No. 2, 3.3 SOP No. 3, and 3.4 SOP No. 4.	44–72
System: Show something "unique" in the CV (Computer Vision).	Introduced a two-part identity rule: a face is accepted only if its match score is at least 0.30 <i>and</i> it leads the next candidate by at least 0.15. In testing, the score alone would have given five strangers a student's queue number; with the margin, none.	66
System: Reduce the 12 algorithms used in the study.	Reduced the algorithms from twelve to seven.	49–59
System: Use appropriate algorithm for facial recognition.	Adopted ArcFace face embeddings through the pretrained InsightFace model, compared by cosine similarity. On held-out photographs, 97.8% of enrolled students were linked, none to a wrong identity, and all 45 non-enrolled probes were refused.	53, 63
System: Refrain from modifying the current queue flow.	Kept the office's existing kiosk and numbering unchanged. Numbers issued by QueueX come from a separate range starting at 5000, so they never collide with kiosk numbers.	53

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System: For students only. Facial recognition: need to register to validate the queue number.	Only students who register their face in the mobile application with an NCF Gbox account, and then ask to be queued, are issued a number by the system.	19, 37
System: Walk-ins (as is printed queue number).	Walk-ins continue to take a printed ticket from the existing kiosk, separately from QueuEx.	19
System: Conduct load testing/pressure testing.	Conducted a load test with 50 concurrent users for two minutes: 2,536 requests with no failures, a median response of 12 ms, and 95% of responses within 31 ms.	70
System: Include how you train your model, not just discuss the algorithm used.	Added the model development: the person-detection and face-recognition models are pretrained, and their decision thresholds were calibrated on 17 people and 45 held-out photographs. The training of the wait-time prediction model is reported under SOP No. 3.	60, 72